

Jose Lima

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EDUCATION

Carnegie Mellon University, Pittsburgh PA

Graduation May 2027

Bachelor of Science in Information Systems | Minor in Computer Science and Game Design

Relevant Coursework: Computer Game Programming, Computer Graphics, Intro to Computer Systems

WORK EXPERIENCE

VR Software Engineer

May - December 2025

Augmented Perception Lab | In-Person

Pittsburgh, PA

- Designed and implemented a motion capture data-collection pipeline in Unity, enabling recording and replay of VR hand movements and OptiTrack motion capture suit data, by programming core systems in C#
- Implemented a three-point calibration system, improving alignment accuracy between OptiTrack skeletons and Meta Quest VR hands, by designing a transformation algorithm to sync coordinate systems
- Developed a Unity-based VR/AR fitness prototype with real-time performance graphs to improve exercise feedback accuracy; presented early prototype to a major footwear company to validate real-world utility

PROJECT EXPERIENCE

Unreal Engine Gameplay Programmer

September 2025 - Present

Independent Work

- Implemented squad-based enemy AI in C++ using Unreal's Behavior Tree framework with custom BTTaskNode and BTDecorator nodes, combining EQS-driven patrol, perception-based combat state transitions, and a NodeMemory-backed strafe task shared across every soldier instance
- Built an interface-driven grappling hook system in C++ around a single IGrappable interface, so the hook launches the player or pulls objects toward the player based entirely on each object's own implementation, with no changes to the hook component needed to add new grappable types
- Designed a data-driven weapon system in C++ where per-weapon tuning, including recoil, spread, and reload timing, lives entirely on UWeaponDataAsset instances, removing weapon-type branching from gameplay code so new weapons ship as a single asset with no code changes

3D Graphics Programmer

August - December 2025

Computer Graphics | Coursework

- Programmed mesh editing operations, enabling geometry manipulation such as edge flip, edge collapse, triangulation, and loop subdivision by extending half-edge mesh data structures in C++
- Built a Monte Carlo path tracer in C++ producing photorealistic global illumination through physically-based materials, including dielectric glass modeled with Fresnel reflectance and Snell's law refraction
- Implemented skeletal animation system in C++ featuring forward/inverse kinematics with gradient descent, automatic bone weight assignment, and linear blend skinning for realistic character deformation

LEADERSHIP EXPERIENCE

Society of Hispanic Professional Engineers (SHPE)

April 2025 - Present

Treasurer

Pittsburgh, PA

- Secured a Purchasing Card, joining a select group of 10 CMU clubs out of 300+ organizations with this financial privilege, improving transparency and efficiency in our chapter spending
- Expanded chapter events by 50% as measured by increased professional, academic, and community programs for members, by securing funding and managing resources effectively

SKILLS

Programming Languages: C++, C, C#, Python

Applications: Unreal Engine, Unity Engine, OptiTrack (Motive), Perforce, Git, Microsoft Visual Studio